

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

COMPUTER COMMUNICATIONS AND NETWORKS

Friday 9 May 2008

14:30 to 16:30

Year 3 Courses

Convener: G Hayes

External Examiners: J Gurd, M Wooldridge

INSTRUCTIONS TO CANDIDATES

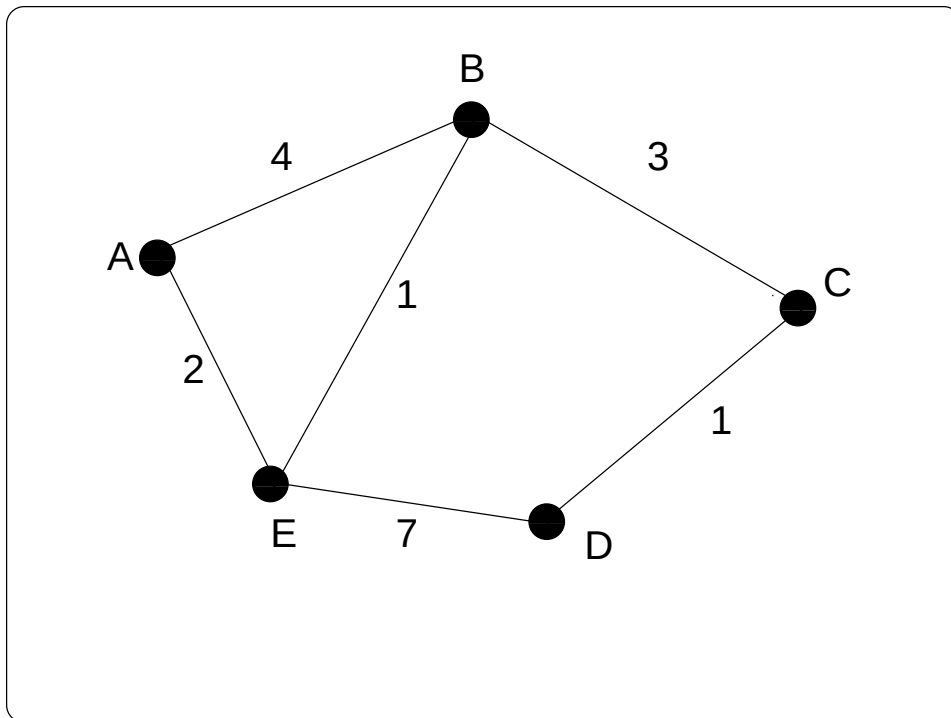
Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY NOT BE USED IN THIS EXAMINATION

1. (a) Define “protocol.” [2 marks]
- (b) Give three advantages of protocol layering. [3 marks]
- (c) Give two main benefits of web caching. Give an example of another application layer protocol that employs a similar caching idea. [3 marks]
- (d) Consider distributing a file of size $F = 10$ Gb to N peers. The server has an upload rate of $u_s = 50$ Mbps, and each peer has a download rate of $d_i = 2$ Mbps and an upload rate of u . For $N = 10$ and 100 , and $u = 250$ Kbps and 500 Kbps, tabulate the minimum distribution times for each of the combinations of N and u for both client-server distribution and P2P distribution. [6 marks]
- (e) Give five reasons why an application developer might choose UDP over TCP as the underlying transport protocol. [5 marks]
- (f) Consider sending a large file from a host to another over a TCP connection that has no loss. Suppose TCP uses AIMD for its congestion control *without* slow start.
 - Assuming *CongWin* increases by 1 MSS (maximum segment size) every time a batch of ACKs is received and assuming approximately constant RTTs (round-trip times), how long does it take for *CongWin* to increase from 1 MSS to 5 MSS (assuming no loss events)?
 - What is the average throughput (in terms of MSS and RTT) for this connection over that period? [4 marks]
- (g) Describe two scenarios where use of standard TCP can result in unfairness. [2 marks]

2. (a) List five key differences between circuit switching and packet switching. [5 marks]
- (b) Datagram networks route each packet as a separate unit, independent of all others. This is not the case with virtual-circuit networks, where each data packet follows a predetermined route. Does this observation mean that virtual-circuit networks cannot route isolated packets from an arbitrary source to an arbitrary destination? Explain. [3 marks]
- (c) What is CIDR (Classless InterDomain Routing)? What is the impact of using CIDR on forwarding? [4 marks]
- (d) Suppose a router has the following CIDR entries in its routing table:
- | Net/Prefix | Next Hop |
|------------------|-------------|
| 128.96.39.0/24 | Interface 0 |
| 128.96.39.128/25 | Interface 1 |
| 128.96.40.0/25 | Router 2 |
| 192.4.153.0/26 | Router 3 |
| default | Router 4 |
- For each of the following IP addresses, what does the router do if a packet with that address arrives? (a) 128.96.39.10; (b) 128.96.40.12; (c) 128.96.39.151; (d) 192.4.153.17; (e) 192.4.153.90. [5 marks]
- (e) The figure shown on the next page indicates the state of the network topology and link costs at node *A* at the end of flooding-based link state dissemination (i.e., using the link-state broadcast algorithm). Determine the shortest path from node *A* to every other node using Dijkstra's shortest path algorithm. Show all steps of the algorithm. [4 marks]
- (f) Give a simple heuristic for finding two paths through a network from a given source to a given destination that can survive the loss of any communication line (assuming two such paths exist and routers never crash). [4 marks]



3. (a) Compare channel partitioning and random access MAC (medium access control) protocols at high and low loads. Which of these two classes of protocols permit statistical multiplexing and why? [7 marks]
- (b) Suppose that the data portion (including the headers) of a packet contains a sequence of 8 bytes, each corresponding to 8-bit unsigned binary representation of the following integers: 243, 81, 128, 64, 32, 16, 27, 9. Compute the Internet checksum for this data. [4 marks]
- (c) Consider the 4-bit generator, $G = 1001$, and suppose that the data portion of the bit stream to be transmitted prior to Cyclic Redundancy Check (CRC) calculation is 11001001. Show the actual bit stream transmitted. Suppose the leftmost bit in the transmitted bit stream is inverted due to noise on the transmission link. Show that this error is detected at the receiver's end. [4 marks]
- (d) An upper-layer packet is split into 5 frames, each of which has an 80 percent chance of arriving undamaged. If no error control is done by the data link protocol, how many times must the message be sent on average to get the entire message through? [6 marks]
- (e) List four differences between routers and switches. [4 marks]